ONLINE MASTERS IN **DATA SCIENCE**

DSC258: NATURAL LANGUAGE PROCESSING

HOW TO DEFINE TERMS

UC San Diego

COMPUTER SCIENCE & ENGINEERING

HALICIOĞLU DATA SCIENCE INSTITUTE

OUTLINE

What's Natural Language Processing (NLP)?

Pre-processing in NLP

- Tokenization
- Stemming & Lemmatization
- **The key: how to define "terms"?**

How to define “term”?

Terms are expected to be orthogonal

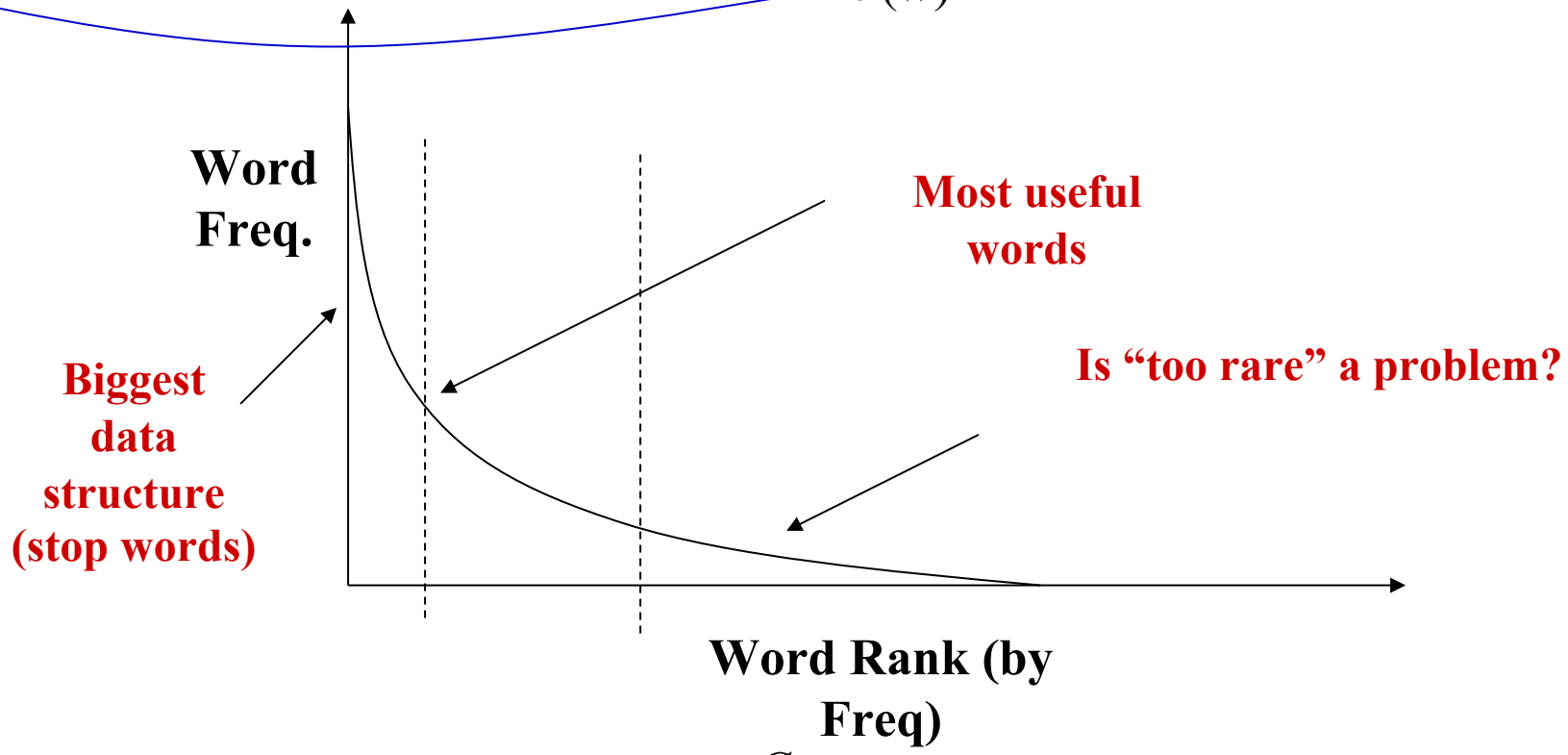
- Ideally “non-overlapping” in semantics

Choices:

- Words
- Stemmed words
- Phrases
- “Latent concepts”
- …

- Rank * Frequency ≈ Constant

$$F(w) = \frac{C}{r(w)^{\alpha}} \qquad \alpha \approx 1, C \approx 0.1$$



Generalized Zipf's law: $F(w) = \frac{C}{[r(w)+B]^{\alpha}}$ **Applicable in many domains**

ZIPF'S LAW – TWO EXTREMES

The most frequent are mostly stopwords

- Stopwords
- The most common words in a language
  - E.g., a, an, the, …
- No single universal list
- We typically assume that stopwords are not useful in NLP
- Can be removed in pre-processing

We typically replace too rare words by a special token "<UNK>"

Will have a closer look in Week 2

Analyzes US news articles on April 9, 2017

Before Phrase Mining

Which “United”?

- United States?
- United Parcel Service?

What’s “Dao”?

- A person?
- A place?

Analyzes US news articles on April 9, 2017

After Phrase Mining

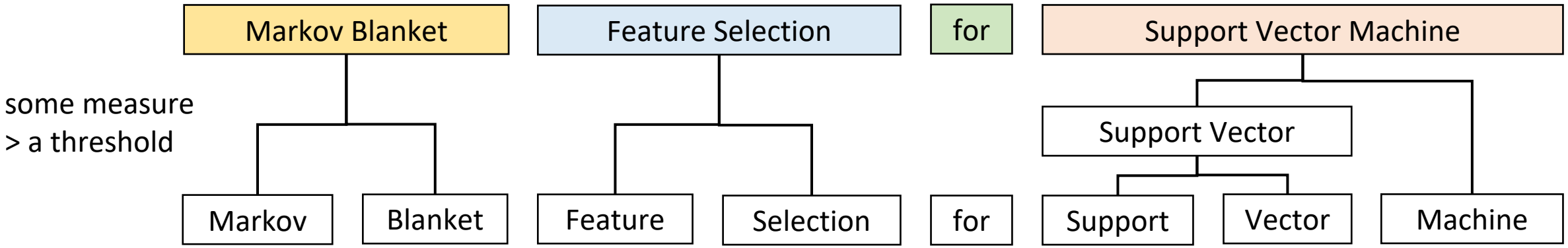
NBC News, April 12, 2017

United Express Flight 3411 incident

- United Airline
- David Dao: A person

Using a greedy merge algorithm!

- Repeat
- Find the most coherent adjacent pairs of “words”
- Merge
- Until the measure is below a threshold



SUMMARY

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